ASE CHAPTER 1

T7 · Apply / Calculate

Productivity and effort — the chapter's only applied relationship — answers in English, notes in Arabic

Rapid self-test — recall prompts (no answers shown)

1. Define software productivity.
2. What is the appropriate unit of effort, and why person-months?
3. Compute LOC/PM for 48,000 LOC at 16 person-months.

Retrieval set — T7 Apply / Calculate

C.1 · Define productivity and state the appropriate unit of effort.

📝 **Answer**

Productivity is **"the rate of output, or production per unit of effort, i.e., the output achieved with regard to the time taken but irrespective of the cost incurred."** Output is measured in **lines of code (LOC)**. The appropriate unit of effort is **"Person Months (PMs), meaning thereby number of persons involved for specified months."**

Aggarwal, §1.5.5, p.14.

C.2 · Apply: a project delivers 30,000 LOC at an effort of 15 person-months. Compute the productivity.

📝 **Answer**

Productivity = output ÷ effort = **30,000 LOC ÷ 15 PM = 2,000 LOC/PM.**

الأرقام (30,000 و15) مثال تدريبي لتطبيق المعادلة؛ أما المعادلة والوحدة فمن المصدر. **Aggarwal, §1.5.5, p.14**

C.3 · Why is productivity expressed as LOC/PM rather than as a raw number of lines?

📝 **Answer**

Because it is a **rate** — output per unit of effort — its unit must combine output and effort, hence **LOC/PM**. A raw line count ignores how much effort produced it.

تفسير الوحدة بوصفها نسبة بين المخرَج والجهد هو توضيح مباشر للمعادلة؛ المصدر يحدد المخرَج والجهد ووحدة الجهد. **Aggarwal, §1.5.5, p.14**